

Lorin Urbantat

Computer Science Student @ ETH Zurich · Software Engineer @ ETH-CES

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Education

ETH Zurich, MSc Computer Science	2025 - 2027
- Expected Graduation: September 2027 - Major: Data Management Systems - Information Security Lab (5.75), Computational Semantics for Natural Language Processing (5.75), Advanced Formal Language Theory (5.5)	
ETH Zurich, BSc Computer Science	2021 - 2024
- Graduated August 2024, GPA 5.04 (6.0 – highest) - Parallel Programming (5.75), Numerical Methods (5.75), Analysis (5.5), Visual Computing (5.75), Applied Computer Architecture (6)	
Waldorfschule Mülheim an der Ruhr, Abitur	2007 - 2020
GPA 1.3 (1.0 – highest)	
Deutsche Schüler Akademie	2019
Summer school for gifted students	

Work Experience

Information Security Lab Teaching Assistant, ETH Zurich	2026 - Present
Test lab exercises and host exercise sessions	
Software Engineer, ETH Chair of Education Systems	2025 - Present
Sole engineer responsible for end-to-end development of a research-focused RAG chatbot using Next.js, Convex and Microsoft Foundry, serving 1,000 users	
Software Engineer, ETH Game Technology Center	2023 - 2025
- Morph Tales Game, Unity 3D - Developed Unity features for Morph Tales and built a game analytics pipeline using Firebase, BigQuery and Looker Studio.	
Coach, mint & pepper	2025
STEM and AI tutor for high school students	
Calculus Teaching Assistant and Exam-Preparation Course, ETH Zurich	2023
Host exercise classes, 15 - 30 students	
Tutor, Waldorfschule Mülheim an der Ruhr	2017 - 2019
Tutorial classes (10 - 25 participants) in Math and Computer Science	

Personal Projects & Relevant Experience

Unikraft on Raspberry Pi 5, ETH Semester Project	2026
- Ported Unikraft to Raspberry Pi 5, implementing AArch64 boot, memory initialization, interrupts and timers - Developed a cache-safe kernel reload mechanism using OpenOCD and GDB, enabling repeated deployment without power-cycling.	
Flateroo, flat sharing platform	2024 - 2025
- React Native application with real-time chat, video streaming infrastructure - Node backend, video transcoding, HLS streaming, PostgreSQL, fly.io	
OpenDemocracy, RAG Powered Voting Advice application for elections	2024
SCION Control Plane Algorithms Test Environment	2024
Built a Docker-based environment and traffic generator to evaluate SCION control-plane algorithms under configurable topologies and changing link conditions.	
TLDR – Lecture Summarizer (VIScon Hackathon)	2023
Lecture summarizer using OpenAI Whisper to create searchable lecture summaries from videos	

Skills & Interests

German (native), English (fluent), French (DELFI B1)	Languages
AArch64 Assembly, C, C++, Kernel Development, TypeScript, React, Python, Java, Git, Linux, Containerization, Unity 3D, React Native, Node.js, SQL, RAG, Machine Learning, PCA	Technologies
Biking, Sports, Cooking, Dancing, Singing	Interests